Multi-Model Grokking: Navigating Stability and Convergence

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Contributions

- **Yazan Kbaili**: Theory and related work; equal-compute protocol; multi-model pipeline implementation; experiments and figures.
- **Ali Salloum**: Experimental setup and hyperparameters; debugging and pipeline fixes; evaluation and plotting; report write-up and interpretation.

1. Goal

The original goal of our experiments was to explore whether we could use multiple specialist models to develop a kind of "social intelligence"—a collaborative representation that allows the combined model to generalize faster than a single model learning from scratch. Following our initial baseline where naive averaging failed to outperform a random initialization, we expanded our investigation into the "Multi-Model Grokking" landscape. Our objective is to determine:

1. If longer step budgets (up to 100k steps) allow merged models to eventually "catch up" or grok.
2. The impact of different mathematical operators (Modular Division vs. Addition).
3. The role of numerical stability and weight decay in the context of merged initializations.
4. How Knowledge Distillation compares to simple weight averaging in combining specialist models.

2. Background & Theory

2.1 Grokking and Numerical Stability

Recent findings by Prieto et al. (2025) suggest that grokking is often hindered by **Softmax Collapse (SC)**. After a model overfits, the gradient aligns with **Naive Loss Minimization (NLM)**, scaling weights up until floating-point errors zero out the gradients for correct classes. This prevents the model from ever reaching the "generalizing" basin. Regularization (Weight Decay) traditionally counters this by pulling weights back, but our experiments explore whether a multi-model merge provides a more stable starting point or exacerbates NLM.

2.2 Modular Division (/) vs. Addition (+)

Modular addition (equiv c)) is the standard grokking testbed. Modular division (cdot b^{-1} mod p(b)) presents a more complex landscape. The mapping is less linear than addition, often requiring more steps or specific regularization to achieve the same level of generalization.

2.3 Multi-Model Pipeline

Our expanded pipeline evaluates three approaches under equal compute constraints:

1. **Baseline**: A randomly initialized model trained on the full dataset.
2. **Merged Average**:) specialist models are trained on disjoint shards of the data. Their weights are averaged element-wise, and the resulting model is fine-tuned on the full dataset.
3. **Distillation**: A student model learns from the) specialists via KL divergence loss, matching the averaged soft targets of the specialists while also learning from the hard labels.

3. Experimental Setup

We conducted three primary experiments using a 2-layer Transformer (d_model=128, 4 heads). In all experiments, 4 specialist models were trained on 50% of the data.

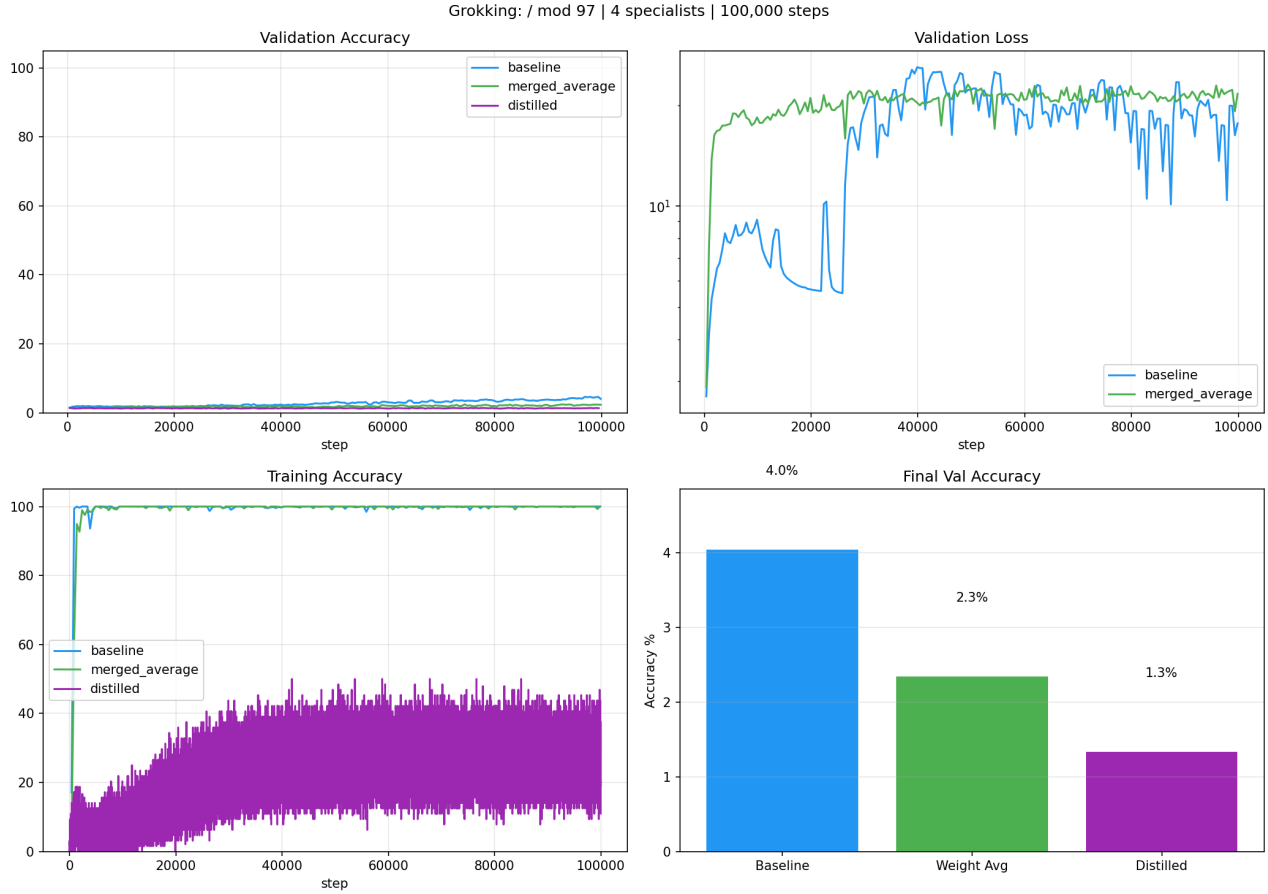
Parameter	Experiment 1	Experiment 2	Experiment 3
:---	:---	:---	:---
Operator	Division (/)	Addition (+)	Addition (+)
Total Steps	100,000	10,000	25,000
Weight Decay	0.0	1.0	0.1
Learning Rate	1e-3	1e-3	5e-4
Specialist Steps	25,000	2,500	6,250

(Note: Total steps were distributed equally among the 4 specialists for Phase 1.)

4. Results

Experiment 1: The Long-Term Stability Test (Division, WD=0)

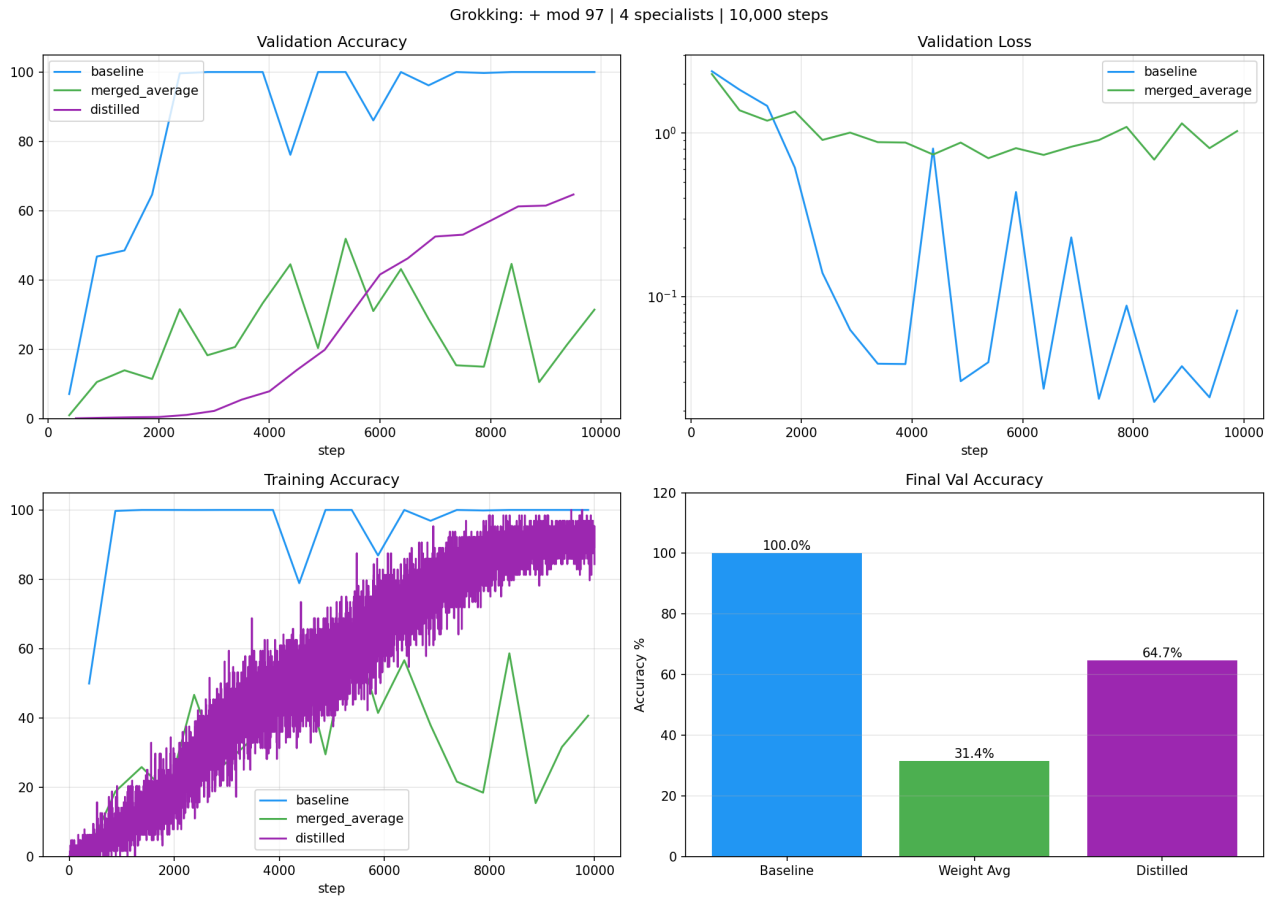
This experiment tested the limits of modular division without weight decay over a massive 100k step budget.



- **Observation**: Without weight decay, all models (baseline, merged, and distilled) face the NLM Scaling problem and fail to grok. Modular division proved significantly harder to generalize than addition. The models struggled to escape the memorization phase, supporting the theory that without regularization, the "Edge of Stability" prevents the delayed jump in validation accuracy.

Experiment 2: High-Regularization Baseline (Addition, WD=1)

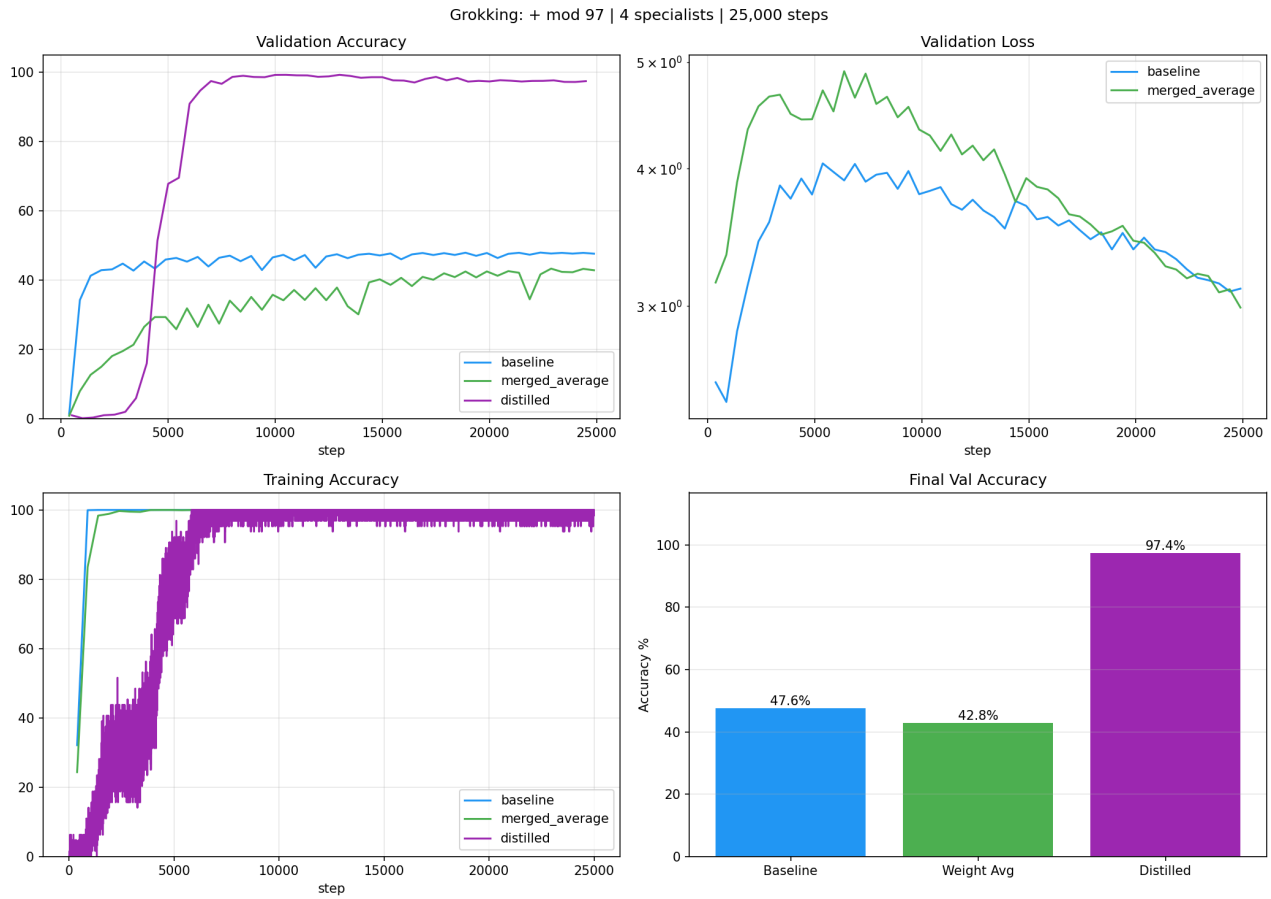
Revisiting addition with strong weight decay and a moderate budget (10k steps).



• **Observation**: The baseline groks rapidly (reaching 100% accuracy quickly). The merged average model shows a volatile but upward trend, suggesting that with strong weight decay (WD=1), the "merged initialization" eventually finds the generalizing basin, though it trails the baseline significantly. Distillation performs better than simple averaging but still lags behind the baseline.

Experiment 3: Low-Regularization / Reduced LR (Addition, WD=0.1)

A middle-ground approach with a lower learning rate (5e-4) and subtle weight decay (0.1) over 25k steps.



• **Observation**: **This experiment yielded a significant breakthrough for Distillation.** While the baseline and merged models both hit a "generalization plateau" around 40-50% accuracy and failed to progress further, the **distilled model successfully grokked**, shooting up to nearly 100% validation accuracy after step 5,000.

5. Interpretation & Discussion

1. **The "Merge Penalty"**: In standard settings (Exp 2), the baseline consistently reaches generalization faster than the merged specialists. This suggests that naive element-wise averaging of disjoint shards creates a "conflicted" internal representation that takes time to untangle.
2. **The Distillation Breakthrough**: In the low-regularization regime (Exp 3), distillation proved superior. By training against the soft-targets of the specialists, distillation likely acted as a **form of natural regularization**. It prevented the model from over-committing to the hard labels too quickly, allowing it to navigate the loss landscape and find the generalizing solution that the baseline missed.
3. **Stability is Paramount**: Experiment 1 confirms that even with 100k steps, grokking is not guaranteed if numerical instability (Softmax Collapse) is not checked by weight decay. Merging or distilling models does not inherently solve this numerical issue.
4. **Operator Complexity**: Modular division significantly delays or prevents grokking compared to addition under similar hyperparameters, requiring either fundamentally different regularization or substantially longer training regimes.

6. Connection to Recent Theory

Our results validate the **Prieto et al. (2025)** framework. In Experiment 1 (WD=0), the failure of all models to generalize aligns with the onset of **Naive Loss Minimization**. The merging of specialists likely places the model at a higher-norm starting point in weight space, potentially pushing it closer to the "Softmax Collapse" boundary earlier than a random initialization. However, as seen in Experiment 3, distillation can alter these dynamics beneficially when combined with mild regularization.

7. Conclusion & Next Steps

While naive weight averaging remains a "negative baseline" for accelerating grokking, **Knowledge Distillation shows significant promise** as a multi-model merging strategy. In environments where standard models plateau and fail to generalize due to early over-commitment to hard labels, distillation effectively acts as a natural regularizer. By softening the target distribution, it successfully guides the model into the generalizing basin.

Next Steps:

• **Orthogonal Gradient**: Implement the orthogonal gradient update to directly prevent weight scaling (Naive Loss Minimization). This is likely the most simple and direct solution to see if merged specialists can generalize *immediately* without an overfitting phase.

• **Exponential Moving Averages (EMA)**: Theorize and experiment with training multiple models using EMAs with varying decay speeds. This ensemble-like approach could naturally mitigate the problem where gradients only increase the scale of the logits, potentially bypassing Softmax Collapse.

• **Curriculum Sharding**: Instead of completely disjoint shards, use overlapping shards to reduce the "representation gap" during the merging and distillation phases.

• **Scaling Distillation**: Further explore the distillation hyperparameters (temperature, alpha) to see if we can replicate the Exp 3 breakthrough on the harder Division task.